Visual Attention & Body Image Disturbance: A Small-Scale Feasibility Test for Tracking Gaze During Body-Oriented Short-Form Video Content Consumption

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Abstract

Body image disturbance (BID) is associated with distinct visual attention patterns while viewing static images of people's bodies; however, it remains unclear whether this observation extends to dynamic media contexts, such as when watching body-oriented short-form video content. As a precursor to a planned between-subjects study comparing the gaze behavior of women with and without BID-driven conditions, we conducted a small-scale feasibility test of our proposed gaze-tracking methodology. Using a Tobii Eye Tracker 5 and a Dell S2421NX monitor, we recruited seven participants to complete directed (instructed fixation on six bodily areas of interest) and undirected (free-viewing) gaze tasks while watching body-oriented short-form videos that varied in body visibility and movement speed. We tested our hardware setup, a calibration validation step introduced partway through testing, and several potential gaze-analysis metrics, including raw gaze, fixation, and dwell-episode hit rates. A motion-robust dwell-episode hit rate, which recognizes sustained gaze on a moving AOI regardless of how the eye is moving, was better suited to these dynamic targets than fixation or pursuit-based metrics (which required per-sample velocity that our 33 Hz hardware could not reliably support). Participants who passed calibration validation tended to attain higher hit rates than those who did not (0.72 vs. 0.53 raw gaze, pooled). Preliminary, hypothesis-generating patterns from the undirected task (n=2) suggest a potential face/head viewing bias, with most fixations falling outside the annotated body regions. We conclude that our pipeline is feasible for the planned full study, contingent on mandatory calibration validation, a revised AOI annotation approach, and continued use of dwell-episode hit rate.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing**; **Interaction techniques**.

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1 Background

1.1 Body Image Disturbance & Visual Attention

Body image disturbance (BID) is a complex, multidimensional psychological phenomenon that expresses as the intense distortion of and preoccupation with one's own physical appearance [8]. Distinct from general aesthetic concern or bodily dissatisfaction, clinically-significant BID is structured around several overlapping dimensions, including cognitive and perceptual, that reinforce the phenomenon over time [1, 5]. Individuals who are acculturated to internalize observers' perspectives of their bodies (a process termed "self-objectification"), notably women and girls, are at increased risk of developing BID [3, 16]. Self-objectification is realized through "habitual body monitoring" and "body surveillance," in which appearance is continually monitored and evaluated against internalized ideals [3]. Sustained by physical comparison [16], this surveillance reinforces dissatisfaction and can lead to compulsive body-checking of one's own body and the bodies of others, as seen in eating disorders from which BID is a clinically-significant feature [18].

Previous work suggests that women with BID-driven conditions, such as Anorexia Nervosa and Bulimia Nervosa, exhibit distinct gaze fixation patterns when viewing images of their own and others' bodies, including increased visual attention to "weight-indicative" areas (e.g., hips, abdomen, thighs) and to areas they perceive as "unattractive" [4, 8–10, 13]. Early research into the connection between visual attention and BID even suggested that gaze tracking may someday be used in the assessment and monitoring of body shape dissatisfaction, given their demonstrated relationship [4].

1.2 Short-Form Video & BID

Short-form video platforms (SFVPs), such as TikTok and Instagram Reels, are social media platforms characterized by vertically oriented, short video content delivered through infinitely scrollable feeds. Their design encourages rapid consumption of videos in succession, or "binge-scrolling," where users passively view large numbers of recommended videos in a single feed for extended periods without seeking out specific content [20]. These platforms are widely used by adolescents and young adults [6], including those in eating-disorder treatment, of which many indicate that SFVPs are their primary or favored social media platforms [17]. SFVPs are saturated with person- and body-centric content. This includes intentionally body-focused posts like workout videos, makeup tutorials, and weight-loss transformations, as well as unintentionally body-centric content such as casual outfit display posts or "get ready with me" (GRWM) routine videos. This constant exposure to body-oriented material has been speculated to impact individuals susceptible to BID. A growing number of papers associate short-form video exposure to bodily dissatisfaction, appearance comparison, and negative mood in nonclinical samples [7, 11];

however, findings remain mixed, and the effects on clinically susceptible populations (as well as the attentional mechanisms underlying them) remain poorly understood. Additionally, distinct from easily-recorded and traditionally-studied social media interactions, including "likes," "comments," and "shares," SFVP use primarily involves passive visual attention during extended scroll periods. This gaze-based mode of engagement, though prevalent, is largely unexplored despite established links between BID, visual attention, and SFVP use in susceptible populations.

2 Goals & Objectives

To understand whether gaze behaviors observed in static contexts translate to dynamic short-form video, we plan to conduct a between-subjects study comparing the gaze patterns of women with and without BID-driven conditions while they watch body-oriented SFVs. If distinct behaviors are observed, this may lay the groundwork for the development of adaptive in-feed interventions. As a precursor, we conducted this small-scale feasibility test with a convenience sample of young women (not screened for BID), evaluating our hardware, gaze-task design, and prospective metrics for quantifying gaze on bodily areas of interest (AOIs) in SFV content varying in body visibility and movement speed.

To our knowledge, the study of gaze on bodily AOIs in BID and BID-driven eating disorders has focused almost exclusively on gaze upon static images [13], virtual reality/avatar-based environments [2], or real-world interactions via mirror exposure [21], as opposed to video-based content, such as body-oriented short-form video. Because established metrics for quantifying gaze on bodily AOIs were developed for static stimuli, it is unclear whether they would be able to transfer to dynamic AOIs that move across the frame. Therefore, evaluating prospective metrics for this setting across body-visibility and movement-speed conditions was a central aim of this feasibility test, alongside testing our overall procedural methodology.

3 Methods

3.1 Video Feed & Annotation

We sourced videos used in our feed from the SnapUGC dataset, a publicly available collection of user-generated SFVs from Snapchat Spotlight [15]. We selected videos featuring a single female-perceived person as the subject of the video without on-screen text to minimize visual distractions. To resemble the variety of body-oriented content encountered on SFVPs, we included videos across a variety of body visibility (full-body, upper-body), movement speed (little/none, slow/infrequent, fast/frequent), and skin visibility (little/none, some, much). We initially screened videos using a combination of YOLOv8 object detection and EasyOCR for text recognition, allowing us to rapidly identify potential candidates before manual screening. We further categorized videos by the visibility of six AOIs. Five (clavicles/collarbones, upper arms, stomach/abdomen/waist, hips, and thighs) were informed by prior static-image gaze-tracking research regarding BID. We additionally included the jawline/neck, reflecting the growing interest toward the jawline in contemporary online appearance culture (e.g., the "looksmaxxing" trend [23]). To ensure consistent viewing conditions, we assembled videos into an auto-scrolling feed where each video played for five seconds before advancing to the next. In the vein of minimizing visual distractions, we omitted on-screen elements such as "like," "comment," and "share" buttons from the feed.

We annotated bodily AOIs within videos using Label Studio (version 1.16.0) by drawing bounding boxes around AOIs at key frames. To create continuous annotations for each frame, we exported the key frame annotations and linearly interpolated the bounding box coordinates across all intermediary frames. This resulted in per-frame bounding box coordinates, which we then used to analyze gaze point containment within each AOI bounding box.

3.2 Participants

We recruited a convenience sample of seven (7) female participants to take part in this feasibility test. All participants were aged 30 or younger (mean: 25, min: 20, max: 30), and most (71%) scrolled on a SFVP several times a day. Two had blue/green eyes (29%), while the remaining participants had brown eyes.

3.3 Gaze Calibration, Validation & Session Flow

We recorded gaze at 33 Hz using a Tobii Eye Tracker 5 (Pupil-Center Corneal Reflection-based), mounted beneath a Dell S2421NX monitor via a custom 3D-printed mount, with participants seated ≈24 inches away. After standard calibration via the Tobii Experience desktop application, the final three participants additionally completed a 7-point calibration validation task (Pass: mean error $\leq 1.0°$; Warn: $\leq 2.0°$; Fail: $>2.0°$, requiring recalibration), introduced mid-study to address gaze-accuracy variability we observed in preliminary testing. Because validation ran before each task, a non-passing result could be addressed by recalibrating and re-validating, making calibration quality an in-session correction rather than a post-hoc diagnostic. In this test, a "Fail" prompted recalibration, whereas a "Warn" was allowed to proceed.

Each session included an undirected (free-viewing) gaze task and a directed gaze task, in which participants were instructed (on-screen and verbally) to fixate on specific AOIs. Calibration and validation were administered at the start of each gaze task, so validation outcomes were session-specific rather than fixed per participant.

We chose the Tobii Eye Tracker 5 for its in-lab availability and prior use in the literature. Although its 33 Hz rate is modest, whether such a low-cost device can still distinguish meaningful gaze patterns on dynamic AOIs was itself part of our feasibility test.

3.4 Gaze-Based Hit Rate Metrics

We adopted three complementary hit-rate metrics: (1) raw gaze, (2) fixation, and (3) dwell-episode.

Raw gaze hit rate was the proportion of gaze samples whose coordinates fell within an AOI's annotated bounding box, among all samples while that AOI was present and annotated. For annotations where two bounding boxes existed in the same frame for the same AOI (e.g., two "upper arm" bounding boxes, each representing a visible arm), a hit in either box was counted as a hit for that AOI. This is a classical AOI-relevant metric that may be adaptable to the dynamic format; however, it is sensitive to saccadic noise.

Fixation hit rate applied the same containment logic to fixations identified via a Dispersion-Threshold Identification (I-DT)

algorithm (40 px / 0.44° visual angle dispersion threshold, 100 ms minimum duration; thresholds selected per Salvucci & Goldberg [19] and confirmed stable across 0.44°–1.33° via sensitivity analysis). A fixation was counted as a hit if its gaze-point sequence centroid fell within the annotated bounding box of the target AOI on the corresponding video frame. I-DT is a classical and commonly-used AOI-relevant metric that is more robust to saccadic noise; however, it was designed for static contexts, where it classifies gaze as a fixation only when it remains in stationary screen coordinates.

Dwell-episode hit rate captured sustained gaze on a moving AOI that screen-space fixation detection may exclude. It is computed over runs of consecutive raw gaze samples within an AOI's bounding box (gap-tolerant up to ≈100 ms, minimum total duration 100 ms), using the same denominator as raw gaze hit rate, so that dwell-episode hit rate ≤ raw gaze hit rate by design.

3.5 Measuring Attention to Moving AOIs

Screen-space fixation detection will underestimate sustained attention to a moving AOI, because gaze that follows a moving target has high screen-space dispersion and is discarded by I-DT as non-fixational. In our case, we were initially interested in whether the behavior it discarded is smooth pursuit, the continuous velocity-matched eye movement that stabilizes a moving target on the fovea. Existing metrics for smooth pursuit generally threshold the eye's momentary velocity, which our 33 Hz Tobii Eye Tracker 5 cannot reliably estimate. This includes velocity-thresholded extensions of I-DT such as I-VDT [14]. Rather than apply these outside their assumptions, we implemented three position-based alternatives to attempt to measure pursuit: two custom pursuit metrics, direction-congruent velocity correlation (whether gaze moved in the AOI's direction) and pursuit gain (whether it matched the AOI's speed), as well as an AOI-relative fixation detector. All three failed. First, direction-congruent velocity correlation was no better than a shuffled baseline. We detected no smooth-pursuit signal, whether because tracking was saccadic or because our short runs and low sampling rate could not reveal one. Second, pursuit gain was unstable as targets slowed (with its speed ratio diverging near zero). Third, the AOI-relative detector, which measured gaze relative to the moving bounding box, injected the box's frame-to-frame annotation jitter (resulting in large part from rapid video cuts and scroll movement classic to SFVs) into the gaze trajectory, fragmenting fixations that were consistent on screen.

Importantly, these results point to a more basic observation: keeping gaze on a moving AOI likely does not require smooth pursuit. Because AOIs are extended regions rather than points, a viewer can sustain attention on a moving body part through a sequence of saccades that sample different parts of it, each landing inside the box, without ever matching the target's velocity. What matters for measuring sustained attention to an AOI is not the oculomotor mechanism but whether gaze remained on the AOI over time. Dwell-episode hit rate asks only whether gaze stayed inside the AOI's bounding box for a sustained interval (a continuous in-box run of at least 100 ms). This is a positional test that requires no velocity estimation, tolerates annotation jitter, and needs no classification of gaze into fixations, saccades, or pursuit. It recognizes sustained visual attention to a moving AOI, however that attention is achieved. As such, we adopted it as our motion-robust hit-rate metric. A tracker with a higher sampling rate, together with an annotation approach that handles rapid cuts and removes scroll-induced motion, could make the velocity-based metrics viable in future work, if distinguishing smooth pursuit from saccadic tracking ever becomes a question of interest.

3.6 Directed vs. Undirected Analyses

For directed-gaze analyses, we stratified hit rates by AOI, movement speed, body visibility, and calibration-validation status, and compared fixation and dwell-episode hit rates by movement speed. Because participants were directed where to look, these analyses assessed the reliability of gaze-based AOI hit-rate measurement with the Tobii Eye Tracker 5 across movement speeds and calibration-validation outcomes. For undirected-gaze analyses (n=2, validation-pass participants only), we examined the same stratifications with skin visibility, the proportion of fixations landing on vs. outside AOIs, and dwell retention (dwell-episode / raw hit rate) by movement speed. Because gaze was unconstrained, these analyses were descriptive and offered a first look at natural gaze behavior during body-oriented SFV viewing.

We report hit rates as sample-weighted pooled rates (total hits divided by total in-window samples) rather than averages of per-participant rates, because sample counts varied across participants and cells, and pooling weights each estimate by the data supporting it.

4 Results

4.1 Directed Gaze

Among the factors we examined, calibration-validation status was associated with the largest differences in hit rates. Validated-pass participants had higher raw gaze (0.715) and fixation (0.818) hit rates than unvalidated (0.525 / 0.587) or the single validated-warn (0.499 / 0.541) participant (Table 1). Because validation was introduced mid-study in a small, non-randomized sample, this is confounded with individual differences and recruitment order, so we interpret it as motivating a mandatory validation check rather than as evidence that validation itself raises accuracy. With validated calibration, this accuracy is encouraging. Even when gaze fell outside the instructed AOI, the median miss was only ∼1° of visual angle from the box edge (0.95° for fixations, 1.03° for raw gaze), implying a near-miss at the calibration tolerance rather than a tracking failure. This suggests the pipeline localizes gaze to roughly the precision needed to distinguish gaze on body regions. Near-miss offsets of this size may also be recoverable with post-hoc drift correction, which we will explore in the full study.

Slow/infrequent movement yielded the highest hit rates in every validation group. Among validated-pass participants, upper-body-framed stimuli (somewhat larger AOIs, ∼1.2×) yielded higher hit rates than full-body-framed stimuli.

Because dwell-episode and raw gaze hit rates share a denominator (all in-window gaze samples), their ratio (dwell retention) cleanly measures how much on-target gaze was sustained rather than momentary. Pooled across participants, dwell retained 98% of raw on-target gaze (0.567 vs. 0.576), and this retention did not

fall when AOIs moved. Rather, it held at approximately 0.98 during stillness and 0.98 to 0.99 during motion (slow and fast pooled) in every validation group. A metric that inherited I-DT's motion penalty would show retention dropping under motion; however, dwell preserved on-target gaze equally well or slightly better when targets moved. The fixation to dwell-episode gap pointed the same way, narrowing during AOI motion (0.050 to 0.099) relative to stillness (0.087 to 0.144) across all three validation groups, although this comparison spans different denominators (fixations vs. raw samples) so we therefore read it qualitatively. Together these support dwell-episode hit rate as a motion-robust complement to fixation for moving AOIs.

4.2 Undirected Gaze (Hypothesis-Generating, n=2)

In our small, exploratory analysis of free-viewing gaze for the two participants who passed calibration validation, both showed their highest raw gaze hit rates on the jawline/neck, which was also highest for fixations pooled across participants (though P2's single highest fixation AOI was the stomach/abdomen/waist), followed by the stomach/abdomen/waist, then thighs (Table 2). However, most identified fixations fell outside any of the six annotated AOIs (P1: 70%, P2: 82%). For both participants, AOI hit rates were higher for full-body than upper-body content across raw gaze, fixations, and absolute fixation counts. One participant's raw gaze AOI hit rate increased with skin visibility (Little/none: 0.096 → Much: 0.117), while the other showed a modest net decrease (0.091 → 0.084).

Pooled across AOIs, dwell retention (the proportion of gaze-AOI overlap reflecting a sustained ≥100 ms dwell rather than a brief pass-through) was highest during fast/frequent movement for both participants (pooled ≈86% vs. ≈80% for the slower conditions). This fast-movement observation in dwell retention held at the AOI level for jawline/neck, upper arms, and stomach/abdomen/waist, but not for thighs, which showed the opposite pattern.

5 Discussion

5.1 Calibration Validation

Calibration-validation status was associated with larger differences in hit rates than any AOI, movement-speed, or body-visibility stratification. Adding the validation made calibration quality actionable as a non-passing result could be addressed on the spot by recalibrating and re-validating. Consistent with this, our one validated-warn participant, who proceeded without recalibrating, had among the lowest hit rates (including a 0.172 AOI cell). This motivates a stricter protocol for the full study in which we recalibrate until we achieve a "Pass," rather than proceed on a "Warn."

5.2 Movement Speed & Dwell Retention

Directed hit rates were highest for slow/infrequent movement, exceeding both little/none and fast/frequent. Given the sample size, we tentatively speculate that a slowly moving target may offer a continuous, low-demand anchor for gaze, whereas prolonged fixation on a static target is vulnerable to attentional and ocular drift [22] and faster motion is harder to follow. Additionally, the dwell-retention result suggests that the dwell-episode metric behaves as intended. On-target gaze was sustained rather than momentary, and remained so when AOIs moved, so the metric captures motion-robust attention. This supports retaining dwell-episode hit rate as a primary metric in the full study.

5.3 AOI Size

Because a larger bounding box is easier for a gaze estimate to fall within, hit-rates depend partly on AOI size, which varied in this feasibility test with both body region and body visibility (full vs. upper-body shots, i.e., body visibility). Differences in hit rate across AOIs therefore reflect box size in addition to attention, so we interpret them cautiously, and plan to normalize for or control bounding-box area and count in the full study.

5.4 Undirected Gaze

The following observations are drawn from the two participants who passed calibration validation (n=2) and are intended as hypothesis-generating rather than population-level claims (note that participants were not screened for BID). The prominence of the jawline/neck across raw gaze and fixation hit rates may reflect a general face/head bias in free viewing, possibly related to social-information-seeking tendencies [12]. The observation that most fixations fell outside any annotated AOI suggests AOI hit rate could be read as relative prioritization among a pre-specified set of regions as opposed to a proxy for overall visual interest in the body. Interpreting it in the full study will require a comparison baseline, such as a non-body control region or a screen-center reference, to establish whether AOI dwell exceeds chance-level allocation across the frame. Higher AOI hit rates for full-body content may reflect more simultaneously visible AOIs, greater overall engagement, or both (as body visibility, AOI count, and AOI size covary here and cannot be separated). The divergent skin-visibility pattern (one participant increasing, one decreasing) may reflect individual differences in response to skin exposure. Finally, the observation that dwell retention peaked during fast movement for jawline/neck, upper arms, and stomach/abdomen/waist, but not thighs, may indicate this is not simply a generic property of high-attention AOIs. A larger, BID-screened sample would clarify to what extent AOI-specific factors (e.g., body region, typical motion) contribute.

5.5 Limitations

This feasibility test used a convenience sample of seven women, split into validation groups of four (no validation), two (passed), and one (warned), which is too small for anything other than descriptive case observations. Because these rates pool gaze samples across participants, they are weighted toward participants who contributed more valid samples and treat within-participant samples as independent. We therefore read them as descriptive aggregates, not inferential estimates. Four participants completed no calibration validation, since this step was introduced mid-way through testing. For the directed task, participant compliance was assumed but not verified, so tracker/calibration error could not be distinguished from non-compliance. Bounding-box annotations exhibited frame-to-frame jitter, including extreme centroid displacements at scroll transitions and rapid cuts, which made AOI-relative fixation detection and frame-level AOI motion-state classification not possible. To address this, we plan a revised labeling strategy for the

Table 1: Pooled hit rate by stratification variable, metric, and calibration validation status (directed gaze task)

Stratification	Metric	No Validation (n=4)	Validated (Pass) (n=2)	Validated (Warn) (n=1)
Overall				
	Raw Gaze	0.525	0.715	0.499
All	Fixation	0.587	0.818	0.541
	dwell-episode	0.514	0.709	0.489
By AOI (body visibility)				
	Raw Gaze	0.499	0.838	0.737
Jawline/Neck (upper body)	Fixation	0.514	0.970	0.800
	dwell-episode	0.481	0.836	0.727
	Raw Gaze	0.538	0.673	0.407
Clavicles (upper body)	Fixation	0.569	0.778	0.506
	dwell-episode	0.529	0.666	0.397
	Raw Gaze	0.413	0.730	0.699
Upper Arms (upper body)	Fixation	0.511	0.807	0.867
	dwell-episode	0.401	0.724	0.685
	Raw Gaze	0.606	0.690	0.172
Stomach/Abdomen/Waist (full body)	Fixation	0.686	0.788	0.159
	dwell-episode	0.597	0.681	0.152
	Raw Gaze	0.452	0.662	0.384
Hips (full body)	Fixation	0.507	0.722	0.446
	dwell-episode	0.442	0.655	0.379
	Raw Gaze	0.638	0.699	0.598
Thighs (full body)	Fixation	0.740	0.841	0.545
	dwell-episode	0.628	0.695	0.597
By Movement Speed				
	Raw Gaze	0.485	0.594	0.508
Little/none	Fixation	0.582	0.728	0.582
	dwell-episode	0.474	0.584	0.495
	Raw Gaze	0.556	0.772	0.509
Slow/infrequent	Fixation	0.660	0.855	0.617
	dwell-episode	0.545	0.770	0.503
	Raw Gaze	0.537	0.697	0.500
Fast/frequent	Fixation	0.546	0.807	0.482
	dwell-episode	0.526	0.688	0.486

Note. Raw gaze and dwell-episode hit rates share a denominator (all in-window gaze samples), so dwell-episode ≤ raw gaze. Fixation hit rate is computed over detected fixations.

full study. Movement speed, body visibility, and skin visibility were naturalistic, time-varying video properties rather than independently manipulated factors, and were likely correlated with each other and with AOI salience as body visibility (full vs. upper) was confounded with AOI identity. With no control condition, patterns such as the jawline/neck prominence cannot be separated from generic free-viewing biases (e.g., screen-center or face/head bias). We also leveraged a consumer-grade Tobii Eye Tracker 5 whose sampling rate is lower than most research-grade eye trackers. Finally, all comparisons came from a single 3-minute auto-scrolling feed and may not generalize to other body-oriented SFV content with different framing, pacing, or body types.

6 Conclusion

This feasibility test indicates that, once a calibration-validation step is introduced, our pipeline can measure whether gaze falls on a commanded body-part AOI with usable accuracy, even from a small sample and a low-cost, consumer-grade eye tracker. The resulting hit rates varied meaningfully with calibration quality, AOI size, and movement rather than as noise. On this basis, we plan to proceed to a full study comparing the gaze patterns of women with and without BID-driven conditions, contingent on three refinements: (1) mandatory calibration validation for all participants, with recalibration until a passing result; (2) continued use of the dwell-episode hit rate as our primary metric for sustained gaze on moving

Table 2: Hit rate by AOI and metric (undirected/free-viewing gaze task, n=2).

Stratification	Metric	P1	P2	Pooled
Overall				
All	Raw Gaze	0.111	0.080	0.096
	Fixation	0.097	0.059	0.077
	dwell-episode	0.096	0.063	0.079
By AOI				
Jawline/Neck	Raw Gaze	0.189	0.111	0.150
	Fixation	0.175	0.068	0.119
	dwell-episode	0.172	0.082	0.126
Clavicles	Raw Gaze	0.083	0.058	0.070
	Fixation	0.067	0.039	0.052
	dwell-episode	0.066	0.039	0.052
Upper Arms	Raw Gaze	0.066	0.058	0.062
	Fixation	0.052	0.032	0.041
	dwell-episode	0.055	0.048	0.051
Stomach/Waist	Raw Gaze	0.119	0.106	0.112
	Fixation	0.125	0.093	0.109
	dwell-episode	0.101	0.089	0.095
Hips	Raw Gaze	0.055	0.062	0.059
	Fixation	0.043	0.057	0.051
	dwell-episode	0.042	0.051	0.046
Thighs	Raw Gaze	0.111	0.068	0.090
	Fixation	0.069	0.067	0.068
	dwell-episode	0.097	0.057	0.077

Note. Raw gaze and dwell-episode hit rates share a denominator (all in-window gaze samples), so dwell-episode ≤ raw gaze. Fixation hit rate is computed over detected fixations. P1 and P2 are the two participants. Pooled combines both ($n = 2$). Movement-speed and skin-visibility breakdowns are summarized in-text.

AOIs while using the Tobii Eye Tracker 5, given its 33 Hz sampling rate; and (3) a revised AOI annotation approach that reduces frame-to-frame bounding-box jitter and improves dwell-episode precision. Beyond these procedural refinements, the exploratory patterns we observed warrant testing in a larger, BID-screened sample beyond our hypothesis-generating n=2. Finally, we could not measure smooth pursuit directly; however, because AOIs are extended regions that gaze can hold through saccadic sampling, sustained attention does not necessarily involve smooth pursuit, so this gap may be inconsequential. A higher-sampling-rate tracker and a revised annotation approach could nonetheless make the velocity-based metrics feasible in future work.

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